Library Book Management and Analysis System

Question:

Write a Python program to manage and analyze a small library's book database using strings, lists, and dictionaries.

Static Input:

Use the following list of strings as input data (**no user input required**):

```
books_data = [
"Inferno,Dan Brown,Thriller,450,4.5",
"Atomic Habits,James Clear,SelfHelp,550,4.8",
"Ikigai,Hector Garcia,SelfHelp,300,4.2",
"Dune,Frank Herbert,ScienceFiction,600,4.6",
"1984,George Orwell,ScienceFiction,500,4.7",
"Sapiens,Yuval Noah Harari,History,650,4.9",
"The Hobbit,J.R.R. Tolkien,Fantasy,480,4.8",
"Becoming,Michelle Obama,Biography,700,4.7",
"The Alchemist,Paulo Coelho,Fiction,350,4.3",
"To Kill a Mockingbird,Harper Lee,Fiction,400,4.6"
]
```

Each string in the list represents **one book's complete information**, with different details separated by commas.

- **Each element (each string) contains 5 pieces of information in this order:**
    1. **Book Title**
    2. **Author Name**
    3. **Genre**
    4. **Price (in rupees)**
    5. **Rating (out of 5)**

Your Task:

1. **Convert** the above list of strings into a **dictionary**, where:

o **Key = Book title**

o **Value = Another dictionary**

o {"Author": ..., "Genre": ..., "Price": ..., "Rating": ...}

The dictionary should look like below:

{'Inferno': {'Author': 'Dan Brown', 'Genre': 'Thriller', 'Price': 450, 'Rating': 4.5}, 'Atomic Habits': {'Author': 'James Clear', 'Genre': 'SelfHelp', 'Price': 550, 'Rating': 4.8}, 'Ikigai': {'Author': 'Hector Garcia', 'Genre': 'SelfHelp', 'Price': 300, 'Rating': 4.2}, 'Dune': {'Author': 'Frank Herbert', 'Genre': 'ScienceFiction', 'Price': 600, 'Rating': 4.6}, '1984': {'Author': 'George Orwell', 'Genre': 'ScienceFiction', 'Price': 500, 'Rating': 4.7}, 'Sapiens': {'Author': 'Yuval Noah Harari', 'Genre': 'History', 'Price': 650, 'Rating': 4.9}, 'The Hobbit': {'Author': 'J.R.R. Tolkien', 'Genre': 'Fantasy', 'Price': 480, 'Rating': 4.8}, 'Becoming': {'Author': 'Michelle Obama', 'Genre': 'Biography', 'Price': 700, 'Rating': 4.7}, 'The Alchemist': {'Author': 'Paulo Coelho', 'Genre': 'Fiction', 'Price': 350, 'Rating': 4.3}, 'To Kill a Mockingbird': {'Author': 'Harper Lee', 'Genre': 'Fiction', 'Price': 400, 'Rating': 4.6}}

2. **Perform the following analyses:**
a. Display the **average price** of all books.
b. Display the **highest-rated book** (title, author, rating).
c. **Group books by genre** and show the **count of books** in each genre.
d. Display all **ScienceFiction** books **sorted by rating (descending)**.
e. Display the **most expensive book in each genre**.

Expected Output:
Average Price of all books: 498.0

Highest Rated Book: Sapiens by Yuval Noah Harari (Rating: 4.9)

Books by Genre:
Thriller: 1
SelfHelp: 2
ScienceFiction: 2
History: 1
Fantasy: 1
Biography: 1
Fiction: 2

ScienceFiction Books (sorted by rating):
1. 1984 — 4.7
2. Dune — 4.6

Most Expensive Book per Genre:
Thriller: Inferno (450)
SelfHelp: Atomic Habits (550)
ScienceFiction: Dune (600)
History: Sapiens (650)
Fantasy: The Hobbit (480)
Biography: Becoming (700)
Fiction: To Kill a Mockingbird (400)

Note: Your program must be a *general solution* that works correctly for any valid input.Hard-coded logic that only works for the sample list will receive **no marks**.Use the list exactly as provided in the question; **do not take user input** in your program.